Assignment 1

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1 Introduction

This is my first LaTeX document. Extra spaces and single line breaks in the source are treated as a single space; a blank line starts a new paragraph. Special characters such as % , $, & , # , _ , { and } must be escaped with a backslash.

1.1 Document class options

The default paper size of the `article` class is `letter`. It is changed to A4 by passing the option `a4paper`, e.g. `\documentclass[12pt,a4paper]{article}`. The font size for the whole document is set the same way (here `12pt`).

1.2 A numbered list

1. first item
2. second item
3. third item

2 Science!

Einstein's mass–energy equivalence is famously written $e = mc^2$ (Einstein, 1905), where e is energy, m is mass and c is the speed of light in a vacuum.

3 Equations

The time-dependent Schrödinger equation (Schrödinger, 1926) is given in Equation (1):

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2\mu}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t) \tag{1}$$

As discussed by Schrödinger (1926), Equation (1) describes how the quantum state Ψ of a particle of reduced mass μ evolves in a potential V.

References

Albert Einstein. Ist die trägheit eines körpers von seinem energieinhalt abhängig? *Annalen der Physik*, 323(13):639–641, 1905. doi: 10.1002/andp.19053231314.

Erwin Schrödinger. Quantisierung als eigenwertproblem (erste mitteilung). *Annalen der Physik*, 384(4):361–376, 1926. doi: 10.1002/andp.19263840404.